

```
if (irrecv.decode(&results)) {  
  
    value = results.value;  
    if (value != 0xFFFFFFFF) {  
        cmd = value;  
    }  
  
    timeStamp = millis();  
  
    irrecv.resume();  
}  
  
timeElapsed = millis() - timeStamp;  
if (timeElapsed < 200) {  
    ... find match here  
}
```

Scenario One

I pressed command once and released it immediately.

the first if records, stores the real signal & time stamp.

time elapsed will remain < 200ms in the first and subsequent loops where matched signal will be performed before 200ms or until new signal detected

Since time stamp is inside new signal detection, as long as no new signal is received before 200ms, time stamp will not be updated to continue the time condition after 200ms

Dry Run for Scenario One: Button is pressed once and released immediately

//remote pressed	0010
if (irrecv.decode(&results)) { //true	0015
	0020
value = results.value; //real remote value	0025
	0030
if (value != 0xFFFFFFFF) { // true	0035
cmd = value; //real remote value stored	0040
}	0045
	0050
timeStamp = millis(); // time marked	0055
	0060
irrecv.resume(); //listen to the next signal	0065
}	0070
	0075
timeElapsed = millis() - timeStamp; // millis = 0080 here	0080
// check how long since last fresh signal	0085
if (timeElapsed < 200) { // true	0090
----- next loop	0095
//remote released	0100
if (irrecv.decode(&results)) { //true	0105
	0110
value = results.value; //noise FFFFFFFF value	0115
	0120
if (value != 0xFFFFFFFF) { // false	0125
cmd = value; // skipped	0130
}	0135
	0140
timeStamp = millis(); // time marked regardless	0145
	0150
irrecv.resume(); //listen to the next signal	0155
}	0160
	0165
timeElapsed = millis() - timeStamp; //millis = 0170 here	0170
// check how long since last fresh signal	0175
if (timeElapsed < 200) { // true	0180
----- next loop	0185
if (irrecv.decode(&results)) { //false	0190
//time update skipped, old time remains	0195
	0200
timeElapsed = millis() - timeStamp; //millis = 200 here	

// check how long since last fresh signal	0205
if (timeElapsed < 200) { // still true for now	0210
-----	0215
	0220
if (irrecv.decode(&results)) { //false	0225
//time update skipped, old time remains	0230
	0235
timeElapsed = millis() - timeStamp; //millis = 240 here	0240
// check how long since last fresh signal	0245
if (timeElapsed < 200) { // still true for now	0250
----- next loop	0255
if (irrecv.decode(&results)) { //false	0260
//time update skipped, old time remains	0265
	0270
timeElapsed = millis() - timeStamp; //millis = 275 here	0275
// check how long since last fresh signal	0280
if (timeElapsed < 200) { // still true for now	0285
----- next loop	0290
if (irrecv.decode(&results)) { //false	0295
//time update skipped, old time remains	0300
	0305
timeElapsed = millis() - timeStamp; //millis = 310 here	0310
// check how long since last fresh signal	0315
if (timeElapsed < 200) { // still true for now	0320
----- next loop	0325
if (irrecv.decode(&results)) { //false	0330
//time update skipped, old time remains	0335
	0340
timeElapsed = millis() - timeStamp; //millis = 345 here	0345
// check how long since last fresh signal	0350
if (timeElapsed < 200) { // false	0355
-----	0360
Both remains false until further signal detected	0365
...	0370
	0375
	0380
	0385
	0390
	0395
	0400
	0405
	0410
	0415

Scenario Two

I pressed command and keep holding it.

the first if records, stores the real signal & time stamp.

time elapsed remain < 200ms in first & subsequent loops where matched signal performed

however FFFFFFFF keeps coming in and time elapsed keeps getting updated so time elapsed remains < 200ms as long as button is held, keeping the running wheels

Dry Run for Scenario One: Button is pressed once and released immediately

//remote pressed	0010
if (irrecv.decode(&results)) { //true	0015
	0020
value = results.value; //real remote value	0025
	0030
if (value != 0xFFFFFFFF) { // true	0035
cmd = value; //real remote value stored	0040
}	0045
	0050
timeStamp = millis(); // time marked	0055
	0060
irrecv.resume(); //listen to the next signal	0065
}	0070
	0075
timeElapsed = millis() - timeStamp; // millis = 0080 here	0080
// check how long since last fresh signal	0085
if (timeElapsed < 200) { // true	0090
----- next loop	0095
//remote is still held	0100
if (irrecv.decode(&results)) { //true	0105
	0110
value = results.value; //noise FFFFFFFF value	0115
	0120
if (value != 0xFFFFFFFF) { // false	0125
cmd = value; // skipped	0130
}	0135
	0140
timeStamp = millis(); // time marked regardless	0145

irrecv.resume(); //listen to the next signal	0150
}	0155
	0160
timeElapsed = millis() - timeStamp; //millis = 0170 here	0165
// check how long since last fresh signal	0170
if (timeElapsed < 200) { // true	0175
----- next loop	0180
//remote is still held	0185
if (irrecv.decode(&results)) { //true	0190
	0195
value = results.value; //noise FFFFFFFF value	0200
	0205
if (value != 0xFFFFFFFF) { // false	0210
cmd = value; // skipped	0215
}	0220
	0225
timeStamp = millis(); // time marked regardless	0230
	0235
irrecv.resume(); //listen to the next signal	0240
}	0245
	0250
timeElapsed = millis() - timeStamp; //millis = 0250 here	0255
// check how long since last fresh signal	0260
if (timeElapsed < 200) { // true	0265
-----	0270
//remote is still held	0275
if (irrecv.decode(&results)) { //true	0280
	0285
value = results.value; //noise FFFFFFFF value	0290
	0295
if (value != 0xFFFFFFFF) { // false	0300
cmd = value; // skipped	0305
}	0310
	0315
timeStamp = millis(); // time marked regardless	0320
	0325
irrecv.resume(); //listen to the next signal	0330
}	0335
	0340
timeElapsed = millis() - timeStamp; //millis = 0340 here	0345
// check how long since last fresh signal	0350
if (timeElapsed < 200) { // true	0355
----- both remains true while button is held	